STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 9: Mechanical Properties of Fluids

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Accept equivalent correct methods.

Suggested mark allocation:

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

Q1 - 1 mark

Normal force per unit area.

Q2 - 1 mark

Pa s.

Q3 - 1 mark

A_1v_1=A_2v_2.

Q4 - 1 mark

p+rhov^2+rhogh=constant.

Q5 - 1 mark

N/m.

Q6 - 1 mark

rhogh.

Q7 - 1 mark

6pirv.

Q8 - 1 mark

2S/r.

Q9 - 2 marks

30,000 Pa.

Q10 - 2 marks

Doubles.

Q11 - 2 marks

Lower.

Q12 - 2 marks

100 N.

Q13 - 2 marks

40 Pa.

Q14 - 2 marks

Bubble has two liquid-air surfaces; Deltap=4S/r.

Q15 - 3 marks

8 m/s.

Q16 - 3 marks

16,000 Pa.

Q17 - 3 marks

approx 1.88x10^-4 N.

Q18 - 3 marks

Hydrostatic balance gives p=p_0+rhogh due to weight of fluid above.

Q19 - 4 marks

4.5 m/s.

Q20 - 4 marks

Deltap=4S/r=60 Pa.